Statistical Methods in Experimental Particle Physics
Scuola Normale Superiore di Pisa

Measurement of D^0 meson lifetime using $D^0 \to K\pi$ decays

Introduction

The measurement of lifetime of particles in HEP is very important to provide information on the fundamental interactions. This is usually an experimental challenge and sometimes experiments are designed and built to specifically to perform this type of measurements. In particular, it is of fundamental importance to have a 'good' spatial resolution, in order to both identify and measure the spatial coordinates of the decay vertex, and a very good momentum resolution. Here, we propose a simple measurement of the D^0 meson lifetime (neglecting any mixing and CP violating effects) using a sample two-body charmless $D^0 \to K^-\pi^+$ decays[1]. The PDG reports the following value for the 'effective' lifetime of the of the D^0 meson into the Cabibbo-Favored and flavor-specific $K^-\pi^+$ final state [1]

$$\tau = (410.3 \pm 1.0) \times 10^{-15}\ \text{s} \quad \Longrightarrow \quad c\tau = 123.01\ \mu\text{m}.$$

The D^0 meson is a pseudo-scalar particle and its mass value is $m_{D^0} = 1864.84 \pm 0.05$ MeV. In the following 'simulated' experiment the value of τ is not equal to its measured value and the experimenter is asked to provide his best estimate, assessing both statistical and systematic uncertainties. An interesting reading on the subject can be found in Ref. [2].

Experimental apparatus and data

Simulated data are produced using a simplified parametric simulation of the LHCb experiment [3]. Curvature, on the xz plane, of individual particles is smeared, as well as the position of the primary vertex (PV) and decay vertex (SV). The LHCb coordinate system is a right-handed system centered in the nominal pp collision point $(0, 0, 0)$, with the z axis pointing along the beam direction towards downstream of the detectors, the y axis pointing vertically upwards, and the x axis pointing in the horizontal direction, outside the center of LHC. The available samples for this experiment are

- `tree_DKPi_data_SB.root` – data (signal plus background): $D^{*+} \to D^0\pi^+ \to [K^-\pi^+]\pi^+$ decays. The true value of lifetime τ is unknown.
- `tree_DKPi_mc.root` – Monte Carlo: $D^{*+} \to D^0\pi^+ \to [K^-\pi^+]\pi^+$ decays ($N_{\text{gen}} = 10^7$ events, $\tau = 410.3$ ps).

Simulated data are collected, as in a real experiment, within the detector acceptance (geometry and tracking) by means of trigger requirements. Acceptance and trigger requirements are reported in Tab. 1. The Monte Carlo sample is generated assuming as input lifetime the PDG value, $\tau = 410.3$ fs. A sample of $N_{\text{gen}} = 10^7$ events are generated for the Monte Carlo decays. The experiment is realistic with $D^{*+} \to D^0\pi^+ \to [K^-\pi^+]\pi^+$ decays saved on a data file, `tree_DKPi_data_SB.root`, which also contains reconstructed candidates from combinatorial background (a combination of random track-particles satifying both acceptance and trigger requirements). The Monte Carlo file `tree_DKPi_mc.root` contains only signal decays with some 'additional' observables displaying the *true* value of the generated quantity.
The following observables are stored in ROOT TTree objects, where the convention $M^0 \to h_1^- h_2^+$ is used for the decay of the neutral meson (M^0 stands for D^0, B^0 or B_s^0):

[1] The inclusion of charge-conjugate processes is implied throughout, except in the discussion where it is necessarily required to distinguish between the two processes.

observable	$D^{*+} \to D^0\pi^+ \to [h^-h^+]\pi_s^+$	units
$IP(h)$	> 60	μm
$p_T(h)$	> 0.8	GeV
$p(h)$	> 5	GeV
$\eta(h)$	> 2	–
$\eta(h)$	< 4.2	–
$p_T(M^0)$	> 2	GeV
$FD_T(M^0)$	–	μm
$p_T(\pi_s)$	> 0.2	GeV
$p(\pi_s)$	> 1.5	GeV

Table 1: Trigger and offline requirements.

- `id` – Only for Monte Carlo decays. For data $id \equiv 1$. For MC $id \equiv 13$ for $D^0 \to K^-\pi^+$ decays.
- `event` – This is empty (999).
- `h1:px,py,pz,pt,p,eta,IP,charge` – Momentum, pseudo-rapidity, impact parameter, and charge of the h_1^- particle.
- `h1:px_true,py_true,pz_true` – True value of momentum components of the h_1^- particle. Only available in the Monte Carlo file.
- `h1:thetaC0,thetaC1,thetaC2,DLL` – Cherenkov angles of $\mathtt{h}_1$ particle for Aerogel,C4F10, and CF4, respectively. Likelihood ratio function $\mathtt{DLL} = \log(\mathcal{L}_K/\mathcal{L}_\pi)$.
- `h2:px,py,pz,pt,p,eta,IP,charge` – Momentum, pseudo-rapidity, impact parameter, and charge of the h_2^+ particle.
- `h2:px_true,py_true,pz_true` – True value of momentum components of the h_2^- particle. Only available in the Monte Carlo file.
- `h2:thetaC0,thetaC1,thetaC2,DLL` – Cherenkov angles of $\mathtt{h}_2$ particle for Aerogel,C4F10, and CF4, respectively. Likelihood ratio function $\mathtt{DLL} = \log(\mathcal{L}_K/\mathcal{L}_\pi)$.
- `piS:px,py,pt,pz,p,eta,IP,charge` – Momentum, pseudo-rapidity, impact parameter, and charge of the π_s particle.
- `PV:x,y,z` – Coordinates of the position of the Primary Vertex.
- `PV:x_true,y_true,z_true` – True values of the coordinates of the position of the Primary Vertex. Only available in the Monte Carlo file.
- `SV:x,y,z` – Coordinates of the position of the Decay (or Secondary) Vertex of the M^0 meson.
- `SV:x_true,y_true,z_true` – True values of coordinates of the position of the Decay (or Secondary) Vertex of the M^0 meson. Only available in the Monte Carlo file.
- `M0:px,py,pt,pz,p,eta` – Momentum and pseudo-rapidity of the M^0 meson.
- `M0:FDx,FDy,FDt,FDz,FD` – Flight distance of the M^0 meson wrt the PV.
- `M0:time` – Proper decay time of the M^0 meson
- `M0:time_true` – True proper decay time of the M^0 meson. Only available in the Monte Carlo file.
- `M0:IP` – Impact parameter of the M^0 meson.

- `M0:Mpipi,MKK,MKpi,MpiK` – Invariant masses calculated with different mass hypotheses.

Momentum, mass and energy are measured in [GeV], length in [meter], time in [second], and angles in [radian]. Pseudorapidity of a given track-particle is defined, as usual, as

$$\eta = -\ln\left[\tan\frac{\theta}{2}\right],$$

where the angle θ is the polar angle (measured in the decay vertex) of the track

$$p_T = p\sin\theta$$
$$p_z = p\cos\theta$$

with

$$p = \sqrt{p_x^2 + p_y^2 + p_z^2}$$
$$p_T = \sqrt{p_x^2 + p_y^2}.$$

The impact parametr IP of a track is defined as the distance between the track trajectory and the PV space position. By reminding the flight distance vector definition

$$\overrightarrow{FD} = \overrightarrow{SV} - \overrightarrow{PV} \quad \text{and} \quad FD = \sqrt{FD_x^2 + FD_y^2 + FD_z^2}$$

the impact parameter of the track (coming from a M^0 decay) can be determined as

$$IP = FD \cdot \sin\alpha,$$

where

$$\cos\alpha = \frac{\overrightarrow{FD}\cdot\vec{p}}{FD\cdot p}.$$

Objectives

The goal of this experiment is to measure the lifetime of the D^0 meson. The candidate is supposed to provide the best estimate of the true value for τ, assigning statistical uncertainty and discussing the most dominat systematic uncertainties.

Suggested procedures and measurements

1. Determine the resolution in measuring the proper decay time using the Monte Carlo sample. Check if the time resolution varies as a function of proper decay time.

2. Determine the acceptance as a function of proper decay time. Use an empirical function to give an analytical description of that.

3. Check if the acceptance changes as a function of the true value of the Monte Carlo lifetime. *Suggestion: reweitght the given Monte Carlo sample to scan different lifetime input values.*

4. Measure D^0 meson lifetime by analyzing the signal+background data (`tree_DKpi_data_signal_SB.root`). *Suggestions: see below.*

5. Discuss the sources of dominant systematic uncertainties. Where it is possibile try to calculate them. *Suggestions: get some inspiration from Ref. [2].*

The convolution between an Exponential function and a Gaussian function is given

$$\begin{aligned} p(t;\tau,\sigma) &= \int_0^{+\infty} \frac{1}{\tau} e^{-\frac{x}{\tau}} \cdot \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(t-x)^2}{\sigma^2}}\, dx \\ &= \frac{1}{2\tau} \exp\left(\frac{\sigma^2 - 2\tau t}{2\tau^2}\right) \cdot \left[\operatorname{erf}\left(\frac{\tau t - \sigma^2}{\sqrt{2}\tau\sigma}\right) + 1\right] \end{aligned}$$

where the Error funtion is defined as

$$\operatorname{erf}(x) = \frac{2}{\sqrt{\pi}} \int_0^x e^{-z^2} dz.$$

References

[1] P.A. Zyla et al. (Particle Data Group), Prog. Theor. Exp. Phys. 2020, 083C01 (2020) and 2021 update. See also https://pdg.lbl.gov/2021/tables/contents_tables.html.

[2] F. Abudinén et al. (Belle II collaboration), Precise measurement of the D^0 and D^+ lifetimes at Belle II, Phys. Rev. Lett. 127 (2021) 211801, arXiv:2108.03216 [hep-ex].

[3] R. Aaij et al. (LHCb collaboration), LHCb Detector Performance, Int. J. Mod. Phys. A 30, 1530022 (2015). arXiv:1412.6352 [hep-ex].